PAULO JOSÉ ARAGÃO JR.

Data, Financial Modeling & AI Engineer | Former CFO | 15+ years in Infrastructure Finance
Florianópolis, Brazil (UTC−3) | Remote-first; open to relocation (visa sponsorship required)
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PROFESSIONAL SUMMARY

Engineer and former CFO who turns operational and financial data into models people decide with, and into the software and AI systems that run on top of them. Fifteen years in infrastructure consulting, from intern to Finance Director: DCF and tariff models for utility concessions, pricing for ~90 tenders a year, budgets, KPIs and BI, and coordination of a World Bank–financed program. Today I build data and AI products end to end: Bayesian forecasting, a financial ERP with 2,000+ automated tests, and LLM agents with guardrails. Used to async work across time zones (full overlap with US Eastern hours) and to explaining numbers to non-technical decision-makers.

CORE SKILLS

- **Data & software:** Python, SQL, PostgreSQL (PL/pgSQL, pgTAP, RLS), Supabase, data pipelines/ETL, REST APIs, FastAPI, TypeScript/Next.js, Docker, CI/CD, Power BI, Excel
- **Modeling & analytics:** financial modeling, DCF and feasibility, concession and tariff modeling, valuation, forecasting (Bayesian hierarchical, TimesFM), backtesting, budgeting and variance analysis, KPIs, cost accounting, pricing
- **AI:** LLM applications and agents, tool calling and MCP, guardrails and human-in-the-loop, testing and evaluation, multi-provider routing (Claude, Gemini, OpenAI), AI-assisted development
- **Leadership & domain:** finance team leadership, project and contract management, stakeholder reporting, ERP and digital transformation; infrastructure and utilities (water, wastewater, drainage, solid waste), project finance, public procurement

PROFESSIONAL EXPERIENCE

Data & AI Engineer — Self-employed (remote)

Florianópolis, Brazil | Feb 2026 – Present

Design, build and operate data, modeling and AI products end to end as sole engineer, with AI-assisted development (Claude Code) and automated tests and CI as the quality gate.

- **CentralOps — AI-native retail ERP.** ~95k lines (Next.js/TypeScript, Python workers, PostgreSQL/Supabase): POS, inventory, double-entry accounting, cash flow and P&L, payroll, Brazilian tax and e-invoicing; 133 migrations, 2,000+ pgTAP tests, CI/CD.
- **Customer-facing LLM agent** behind an MCP server (16 tools): acts only through database functions under signed context tokens and row-level security; a guardrail checks every price and stock figure against the database; human handoff when unsure.
- **Election-night projection model (Brazil 2026):** Bayesian hierarchical model on official electoral data (national → regional → state shrinkage); uncertainty calibrated by replaying the 2014, 2018 and 2022 runoffs in real vote-arrival order; 56 tests.

- **Trading Copilot (B3):** signal tracking, broker-statement reconciliation of real costs and slippage, capital-gains tax, strategy backtests with TimesFM forecasts; 100+ tests.
- Also: a multi-tenant WhatsApp AI agent (300+ tests), a career-automation platform (FastAPI, React, Chrome extension), a sports-pool SaaS (Next.js, Expo, Stripe) and a public-tender classifier; all self-hosted on Docker/Coolify.

Independent Consultant — Infrastructure Project Finance & Modeling

Florianópolis, Brazil | Feb 2026 – Present

- Advise large Brazilian corporate groups (under NDA) on major infrastructure projects: engineering-financial models (CAPEX/OPEX, cash flow, feasibility) for solid-waste infrastructure and coordination of sanitation engineering projects.

MPB Engineering — Florianópolis, Brazil

Infrastructure engineering consultancy | May 2008 – Jan 2026

Finance Director (CFO) | Feb 2023 – Jan 2026

- Led a 5-person finance team; introduced the annual budget with monthly variance tracking and the KPIs used to monitor operational goals.
- Built BI reporting on department, project and major-expense performance; redesigned ERP cost accounting with cost centers and timesheet-based labor allocation.
- Automated reporting and document workflows with n8n, LLM APIs and Power Platform integrated with the ERP and SharePoint, eliminating manual data entry.

Senior Engineer — Feasibility Modeling, Proposals & Projects | Aug 2011 – Jan 2023

- Built DCF feasibility and tariff models for municipal water and wastewater concessions; analyzed concession processes for four municipalities.
- Prepared financial proposals for the firm's ~90 tenders a year (QCBS/LCS) and designed a metrics-based pricing method (type, size, risk) that significantly cut bid-preparation time.
- Managed 5–7 concurrent contracts (~US$4M a year) supporting infrastructure investments of US$20–150M; owned scope, schedule, budget, risk and client reporting.
- General coordinator and engineer of record for the **World Bank–financed Lagoas do Norte Program** in Teresina (2014–2019): integrated design of sewerage for 312,000 residents, 28 km of drainage, water, roads, a 1 MW biogas plant and BIM urban design across ~1,300 ha.
- Engineer of record on 46 registered engagements, including a river-basin plan (99,000 km², 2.1M people), water systems for state utilities (up to 1,250 L/s) and environmental management of highway programs (BR-470 for DNIT; Arteris concession, 357 km).
- Led the firm's ERP, SharePoint, ISO 9001 and BIM rollouts with in-house staff.

Engineering Intern | May 2008 – Aug 2011 — financial proposals, wastewater design, cost estimates and technical reports.

EDUCATION & CREDENTIALS

- **B.Eng., Sanitary and Environmental Engineering** — UFSC, Brazil, 2011
- **MBA, Business Management** — Fundação Getulio Vargas (FGV), 2015
- **MBA, Building Information Modeling** — Universidade Paulista (UNIP), 2021
- **Registered Engineer** — CREA-SC, Brazil (SC 108928-7)

Languages: Portuguese (native) | English (fluent, C1; lived and worked in the US) | Spanish (B1)